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DEPARTMENT OF COMPUTER SCIENCE

IT3708 - BIO-INSPIRED ARTIFICIAL INTELLIGENCE

Understanding Optimization Landscapes using Bio-inspired Algorithms

Project 3

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1 Introduction

Optimization plays a key role in machine learning, particularly in feature selection, where the goal is to identify a subset of features that maximizes predictive performance while maintaining simplicity. The difficulty of this task is largely determined by the structure of the fitness landscape, which governs how accessible optimal solutions are to a given search algorithm. In this project, three bio-inspired optimization methods, a single-objective genetic algorithm, a multi-objective evolutionary algorithm, and a swarm intelligence algorithm, are applied to both real-world and synthetic landscapes to evaluate how algorithm design influences exploration behavior and solution quality.

2 Problem Instances

Five benchmark landscapes were used for training and evaluation. Three were derived from real-world classification datasets, one was synthetic, and two additional datasets were provided closer to the deadline for final evaluation. Each landscape encodes classification accuracies obtained by training machine learning models on all possible feature subsets, with each index corresponding to the decimal representation of a binary feature selection vector.

The first landscape was constructed from the breast cancer diagnostic dataset of Wolberg 1990, comprising 9 numerical features and a binary target variable. The second was derived from the letter recognition dataset of Slate 1991, which represents 26 capital letters rendered across 20 fonts with added noise, described by 16 features and a 26-class target. The third originated from the credit card application database of Quinlan 1987, containing 15 anonymised features and a categorical target.

The fourth landscape was a synthetic construction proposed by Sánchez-Díaz and Mengshoel 2024, in which a function maps binary strings to integer fitness values in $[0, 4]$ based on the number of active bits. The landscape is parameterised by n , m , and s , set to 16, 1, and 4 respectively, yielding a 16-bit search space.

3 Methodology

3.1 Reading the training data

The training data were contained in HDF5 files where each solution to the 3 problems were contained along side the accuracy of that combination of features, and the training time. The first task was to parse this into accessible files and a format that we could understand. The chosen representation which seemed easy to implement and human-readable was CSV-files. Afterwards, the synthetic problem dataset was created by computing all the fitness values of the triangle landscape.

3.2 Fitness calculations

As mentioned in the previous section, each problem consisted of a dataset containing accuracies from a machine learning model. Each row represented a combination of features, and each column was a separate training run. The design of the fitness function for these problems should, according to the project description, be the average of these accuracies.

Generally, in feature selection problems, it is preferable to have fewer features if possible. For this reason, a penalty was needed to prevent excessive feature use. The penalty function chosen was in regard to how many active bits in a given solution an computed with the formula 1. Where the k is the number of active bits and N is the number of total bits. This is then multiplied with the ϵ noted as the penalty constant.

$$Penalty = \frac{k}{N} \cdot \epsilon \quad (1)$$

In addition to the datasets, the synthetic triangle landscape from Sánchez-Díaz and Mengshoel 2024 was implemented and explored. The fitness function for this landscape was implemented as it was defined in the article. For the testing version, a list of fitness values was provided to simplify the calculations.

3.3 Visualization

After ingesting all training sets and constructing the synthetic dataset, we proceeded to visualize the corresponding fitness landscapes. The chosen visualization technique was a Hinged Bit-Map, in which the two axes represent the two halves of the bit string and each point denotes a candidate solution, as shown in Figure 1. To enhance interpret-ability, we additionally constructed a three-dimensional representation, where the height of each point encodes the fitness associated with the corresponding bit-string configuration.

We chose to visualize 3 key aspects of how the algorithms worked, namely diversity over time, fitness distribution, and niching behavior. Diversity was calculated as entropy over time, using equation 2, plotted against fitness. Fitness distribution was illustrated by plotting a histogram showing the distribution of the landscape compared to the final generation of the population. To illustrate how the algorithms evolve over generations, we implemented an approach that logs the population at each generation and projects these data onto the precomputed landscape, enabling an animated visualization of the search dynamics. For this dynamic visualization, we again employed a Hinged Bit-Map representation. However, to reduce computational complexity, the three-dimensional surface was omitted and replaced with a two-dimensional heatmap, depicting the fitness values and number of individuals in a given region of the landscape.

$$H = - \sum_i p_i \log_2 p_i \quad (2)$$

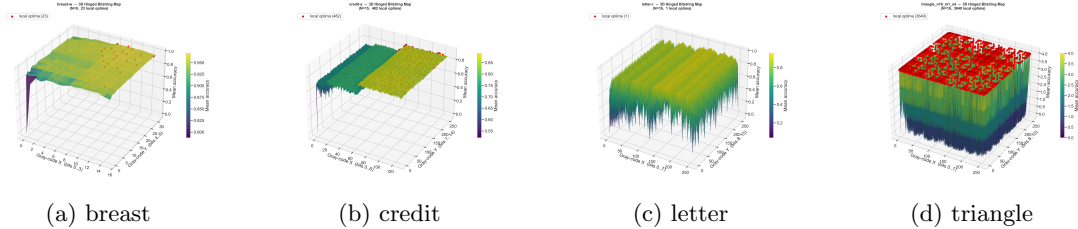


Figure 1: 3D Hinged Bit-Maps of the breast, credit, and letter datasets

3.4 Algorithm Implementation

Three stochastic search algorithms are evaluated on four binary fitness landscapes. Three landscapes are derived from HDF5 files storing cross-validated classifier performance for every non-empty feature subset, where each bit pattern maps to a mean classification accuracy and mean evaluation time aggregated across all stored instances. The fourth landscape is a synthetic asymmetric triangle function with structural parameters fixed according to the project specification. All landscapes incorporate a scalar penalty proportional to the fraction of active features, controlled by a coefficient ϵ held constant across algorithms within each experimental condition.

The first algorithm is NSGA-II, configured to maximise accuracy while simultaneously minimising either evaluation time or feature count depending on the landscape. It employs tournament selection, simulated binary crossover, and bit-flip mutation with an adaptive rate governed by population diversity. The second algorithm is a Binary Particle Swarm Optimiser (BPSO), in which

each particle maintains a real-valued velocity updated via inertia, cognitive, and social components, mapped to bit-flip probabilities through a bounded transfer function with a scheduled mutation rate that decays over the run. The third algorithm is a single-objective genetic algorithm using tournament selection, elitism, and bit-flip mutation. Both BPSO and the single-objective GA optimise mean classification accuracy minus the feature penalty. Population size scales quadratically with problem dimensionality for NSGA-II and BPSO, while the GA uses a fixed population of 200. All runs are explicitly seeded, with independent runs differing only in their random seed while all hyperparameters remain constant, supporting reporting of mean and standard deviation of best fitness across runs.

Table 1: Hyperparameters for each algorithm with default values

NSGA-II	Value	Binary PSO	Value	Single-obj. GA	Value
Population size	$2N^2$	Swarm size	$2N^2$	Population size	200
Generations	100	Iterations	100	Generations	50
Crossover rate	0.9	Inertia w	$0.9 \rightarrow 0.4$	Crossover rate	0.05
Mutation rate	$1/N$	Cognitive c_1	2.0	Mutation rate	0.05
Diversity low	0.25	Social c_2	2.0	Elites	2
Diversity high	0.50	Velocity clamp	6.0	Tournament k	3
Mutation boost	$1.5 \times$	Mutation rate	$1/N \rightarrow 0$	Crossover points	1
Mutation decay	$0.95 \times$				
Mutation max	0.5				
(a) NSGA-II		(b) Binary PSO		(c) Single-objective GA	

4 Results

4.1 Results From Training Instances

The tables below summarize the performance of our single-objective genetic, Non-dominant Sorting Algorithm and the Particle Swarm Optimization algorithm across the four training landscapes. Each row aggregates 90 independent runs (18 hyperparameter configurations $\times$ 5 runs each), reporting the mean and standard deviation of the best accuracy found per run, along with the single best solution discovered across all runs and the known global optimum for reference.

As seen in Table 2, the single-objective genetic algorithm consistently reaches the global optimum on the Triangle and Letter landscapes, with zero variance across runs indicating that the found solution was found in every single run. The Breast landscape shows very tight clustering just below the global optimum, while Credit exhibits the largest gap and the highest variance. This indicates that the Credit landscape is the most difficult of the four for our single objective algorithm to navigate.

Landscape	Average accuracy	Std deviation	Best found accuracy	Best found gene	Global optimum
Triangle	4.0	0.000000	4.0	0001010000100001	4.0
Breast	0.966304	0.000032	0.966311	100001010	0.970122
Credit	0.883668	0.004666	0.887117	000100001100011	0.891263
Letter	0.956333	0.000000	0.956333	0000011111111111	0.956333

Table 2: Results of single-objective GA, using a variety of hyperparameter. Bit-strings are in big-endian

Table 3 shows the results of NSGA-II, which was able to consistently find the global optima for each problem. Credit was the only landscape with a non-zero standard deviation.

Landscape	Average accuracy	Std deviation	Best found accuracy	Best found gene	Global optimum
Triangle	4.0	0.000000	4.0	1010000000010100	4.0
Breast	0.970122	0.000000	0.970122	111001111	0.970122
Credit	0.891253	0.000055	0.891263	001110001110111	0.891263
Letter	0.956333	0.000000	0.956333	000001111111111	0.956333

Table 3: Results of Multi-objective NSGA-II, using a variety of hyperparameter. Bit-strings are in big-endian

Finally, Table 4 shows the performance of our PSO implementation. Here, the algorithm only reaches the global optimum in the triangle landscape, while falling short in the rest. The standard deviations are 0 for all the landscapes, indicating that the algorithm is consistent in its results.

Landscape	Average accuracy	Std deviation	Best found accuracy	Best found gene	Global optimum
Triangle	4.0	0.000000	4.0	0001010000011000	4.0
Breast	0.955640	0.000000	0.955640	010001000	0.970122
Credit	0.871173	0.000000	0.871173	000000101000000	0.891263
Letter	0.947344	0.000000	0.947344	0000000111111111	0.956333

Table 4: Results of Binary Particle Swarm Optimization, using a variety of hyperparameter. Bit-strings are in big-endian

4.2 Visualizations

Figure 2 below show the different aspects of the single-objective genetic algorithm, as defined in section 3.3. Entropy, shown in Figure 2a, declines rapidly at the start of the algorithm, and stays at a consistent level afterwards. The fitness distribution in Figure 2b shows that the landscape has a majority of its datapoints around the areas of higher fitness, with fewer points that have lower fitness. This is consistent with Figure 1a, which shows that the fitness landscape is very flat, with most of the points being on the higher end of fitness. The algorithm appears to prefer these higher-fitness areas, although it maintains a population in the lower-fitness regions that cover a larger area. Finally, Figure 2c shows a heatmap of the population distribution over the state space. Here we can also see a similar trend to before, with most of the individuals centering around the higher-fitness areas while also maintaining a population in a separate low-fitness region.

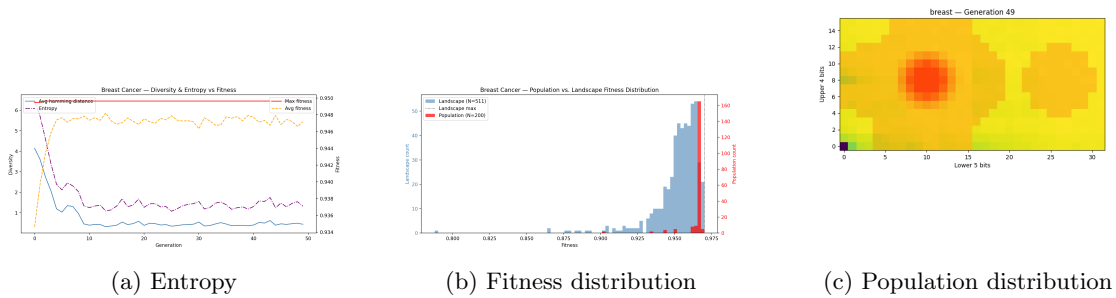


Figure 2: Breast landscape visualizations

4.3 Results From Test Instances

Table 5 summarizes the performance of NSGA-II across the three test landscapes, aggregating 30 independent runs. The algorithm reaches the global optimum on both landscapes, with an average accuracy of 0.979 for Zoo and 0.934 for Hepatitis.

Landscape	Average accuracy	Std deviation	Best found accuracy	Best found gene
Zoo	0.979063	0.000571	0.979167	0101010011001001
Hepatitis	0.933941	0.003188	0.937500	0101101000000000010
Asym. Triangle	5.000000	0.000000	5.000000	00100001100100000010000000000000

Table 5: Results of Multi-objective NSGA-II on test instances (30 runs). Bit-strings are in big-endian

4.4 Results on Synthetic Deceptive Problem

The Asymmetric Triangle proves to be the hardest landscape: despite having a global optimum of 6.0 at popcount 31 (the all-ones bitstring), NSGA-II converges to the local peak at accuracy 5.0 (popcount 5, 15 or 25) in every run.

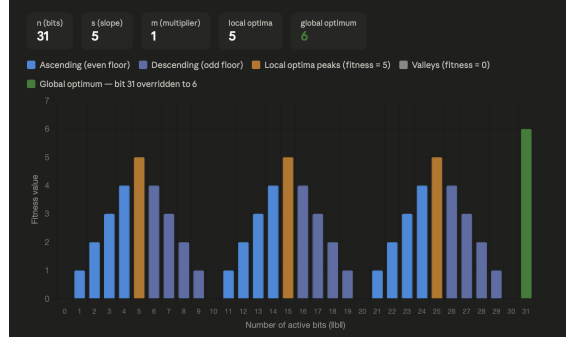


Figure 3: Showcase of the different Fitness values for active bits in the synthetic problem

5 Discussion

5.1 Why NSGA-II Worked?

NSGA-II proved to be the most effective algorithm for locating global optima across the test instances, primarily due to its multi-objective Pareto-based search strategy. On the Zoo landscape ($N = 16$), NSGA-II achieved an average accuracy of 0.979, virtually matching the global optimum of 0.9792 with a standard deviation of just 0.0006. This advantage stems from NSGA-II’s simultaneous optimization of accuracy and inference time, which forces the population to maintain a diverse spread along the Pareto front rather than collapsing prematurely onto a single solution. This diversity acts as an implicit exploration mechanism, by preserving trade-off solutions across different feature counts. The algorithm effectively searches multiple regions of the fitness landscape in parallel, increasing the probability of encountering the global optimum.

On the Hepatitis landscape ($N = 19$), NSGA-II successfully identified the global optimum (0.9375) in its best run, demonstrating its ability to reach the true peak even in a larger search space of over 500,000 candidate solutions. NSGA-II’s crowding distance mechanism and non-dominated sorting provide a principled balance between exploration and exploitation that generalises well across varying landscape topologies.

5.2 Why Did It Not Work On The Synthetic Problem?

The algorithm failed to converge on the global optimum in the synthetic problem, attributable to the inherently deceptive nature of the fitness landscape. Through the nature of the implementation of the NSGA-II algorithm could not the global optimum be found, as the algorithms seeks to minimize the number of active bits as its second objective. The landscape exhibits a repetitive triangular structure with pronounced local optima at fitness 5, providing strong selective pressure toward these suboptimal solutions. The global optimum is only reachable by traversing the final valley at bit position 30 and activating all 31 bits, a region of the search space that offers no gradient signal guiding the population toward it.

The extreme scarcity of the global optimum further compounds this difficulty. Only a single bitstring, $2^{31} - 1$, achieves the global maximum out of $2^{31} \approx 2.1 \times 10^9$ candidate solutions, rendering stochastic discovery highly improbable. Furthermore, reaching the optimum from any high-fitness solution in the vicinity of the local optima requires simultaneously flipping multiple bits, meaning that standard single-bit mutation operators are insufficient to bridge the gap between local and global optima.

Several modifications could be introduced to improve the algorithm’s ability to locate the global optimum. One avenue is increasing population diversity, of which multiple implementations exist. The algorithm employed here incorporated an adaptive mutation rate multiplier, which activated when population diversity (measured by pairwise individual similarity) fell below a threshold of 25%. An alternative strategy would be to implement a partial population restart upon stagnation detection, initializing a subset of individuals at unexplored regions of the search space, thereby increasing the likelihood of discovering niches in proximity to the global optimum. A more straightforward approach would be to scale the population size or the number of generations, as a larger or longer-running search increases the cumulative probability of sampling the global optimum through exploration.

6 Conclusion

This project compared a single-objective GA, NSGA-II, and PSO across real and synthetic feature selection landscapes. NSGA-II performed best overall, reaching or nearly reaching the global optimum on every real-world dataset, as its Pareto-based search preserved diversity and explored multiple regions in parallel. The single-objective GA was competitive on smoother landscapes but struggled on Credit, while PSO converged reliably but to clearly suboptimal solutions.

The Asymmetric Triangle exposed a key weakness, when the secondary objective conflicts with the location of the global optimum, multi-objective search is actively misled. Combined with the landscape’s deceptive structure and lack of gradient signal, this drove every run to a local peak. The takeaway is that landscape structure should inform algorithm and objective choice, and that diversity-preserving mechanisms or adaptive restarts are likely needed when deception is suspected.

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